List of Names:

- TVS,topological vector space;
- LCTVS, Local Complact Topological Space;

Chapter 1 Topological Vector Space

Premilinaries:

- General Topology;
- Filter;
- Linear Space, Linear Maps;

1.1 Definition

1.1.1 Linear Structure, and Compatibilities

Definition 1.1.1.1 Let E be a vector space over the field of complex field F. Let

$$A_v : E \times E \to E \qquad (x, y) \mapsto x + y$$
$$M_s : F \times E \to E \qquad (\lambda, x) \mapsto \lambda x$$

Then, a topology τ on E is said to be **compatible with linear structure of** E if:

- $A_v : (E \times E, \tau \times \tau) \to (E, \tau)$ is continuous under product topology;
- $M_s : (F \times E, \mathcal{F} \times \tau) \to (E, \tau)$ is continuous under product topology.

Then,

Definition 1.1.1.2 A vector space compactible with linear structure of E is called **topological vector space**(TVS).

Remark The topology is 'translation invariant'.

1.1.2 title

Definition 1.1.2.1

- (Absorbing)
- (Balanced)

▶ ***Theorem*** A filter $\mathcal{F}$ on the vector space E is the filter of neighborhoods of the origin in topology compatible with the linear structure if and only if it has the following properties:

- The origin belongs to every subset U belonging to $\mathcal{F}$;
- To every $U \in \mathcal{F}$
-
-
-

∘ **Proof** ... □

1.2 Hausdorff Topological Vector Spaces

Definition 1.2.0.1 ...

1.3 Quotient Topological Vector Spaces

1.4 Cauchy Filter, Complete Subsets, Completion

1.5 Compact Sets

Chapter 2 Semi-Norms,Norms

2.1 Semi-Norms

2.1.1 Convexity

Definition 2.1.1.1 ...

2.1.2 Barrel TVS

Definition 2.1.2.1 ...

2.1.3 LCTVS

Local Convexity

Definition 2.1.3.1 A TVS is said to be locally convex space if there is a basis of neighborhoods in E consisting of convex sets.

♠ **Proposition 2.1.3.2** ...

∘ **Proof** ... □

2.2 Metrizabible TVS

2.3 Norms

2.3.1 Compactness

Riesz's Lemma

▶ ***Theorem*** TFAE: X be a normed space, then

- $\bar{B}_X(0,1)$ is compact;
- $dim(X) < \infty$;

∘ **Proof** ... □

To show the property of compactness

Chapter 3 Frechet Space

3.1 Definitions

3.1.1 F-Space

Definition 3.1.1.1 A **Frechet space** is a TVS with following properties:

- It is metrizable;
- Compelete;
- Locally convex;

Sometimes written as F-Space

♠ **Proposition 3.1.1.2** The following propositions are equivalent:

- (E, P) is Frechet space;
- (E, P) is a LCTVS, and P can be choosen as a countable family.

∘ **Proof**

- $\Rightarrow$
- $\Leftarrow$ Metric

$$d(x, y) := \sum_{j=0}^{\infty} 2^{-j} \frac{P(x, y)}{1 + P(x, y)}$$

□

3.1.2 LF-Space

Definition 3.1.2.1 ...

Compeleteness

◇ **Lemma 3.1.2.2** Suppose E be LF-Space, with filtration

$$E_0 \subseteq \cdots \subseteq E_n \subseteq E_{n+1} \subseteq \ldots, E = \cup E_n$$

We have: For sequencee $(x_n) \in Seq(E)$,

(x_n) converges$\Leftrightarrow$ There exists some $N \in \mathbb{N}$ such that: $\forall x_n \in E_N$, (x_n) converges in E_N.

∘ **Proof** ... □

♠ **Proposition 3.1.2.3** ...

∘ **Proof** ... □

3.1.3 Hormander's Proposition for $C_c^\infty(\Omega)$

◇ **Lemma 3.1.3.1** ...

∘ **Proof** ... □

▶ ***Theorem*** (Hormander) Suppose $\Omega \subseteq \mathbb{R}^n$. We have

$$C_C^\infty(\Omega) = \lim_{K \subseteq \Omega, K \text{ is compact}} C_c^\infty(K)$$

∘ **Proof** ... □

3.2 The Space of Holomorphic Functions in Open Subset of $\mathbb{C}^n$

3.3 The Space of Formal Power Series in n Interminates

3.4 LF Space

Chapter 4 Normable Space, Banach Space

4.1 Norms

4.2 Linear Operators on a Banach Space

4.2.1 The Hille-Yosida Theorem

Chapter 5 Lesbegue Space

Chapter 6 Sobolev Spaces